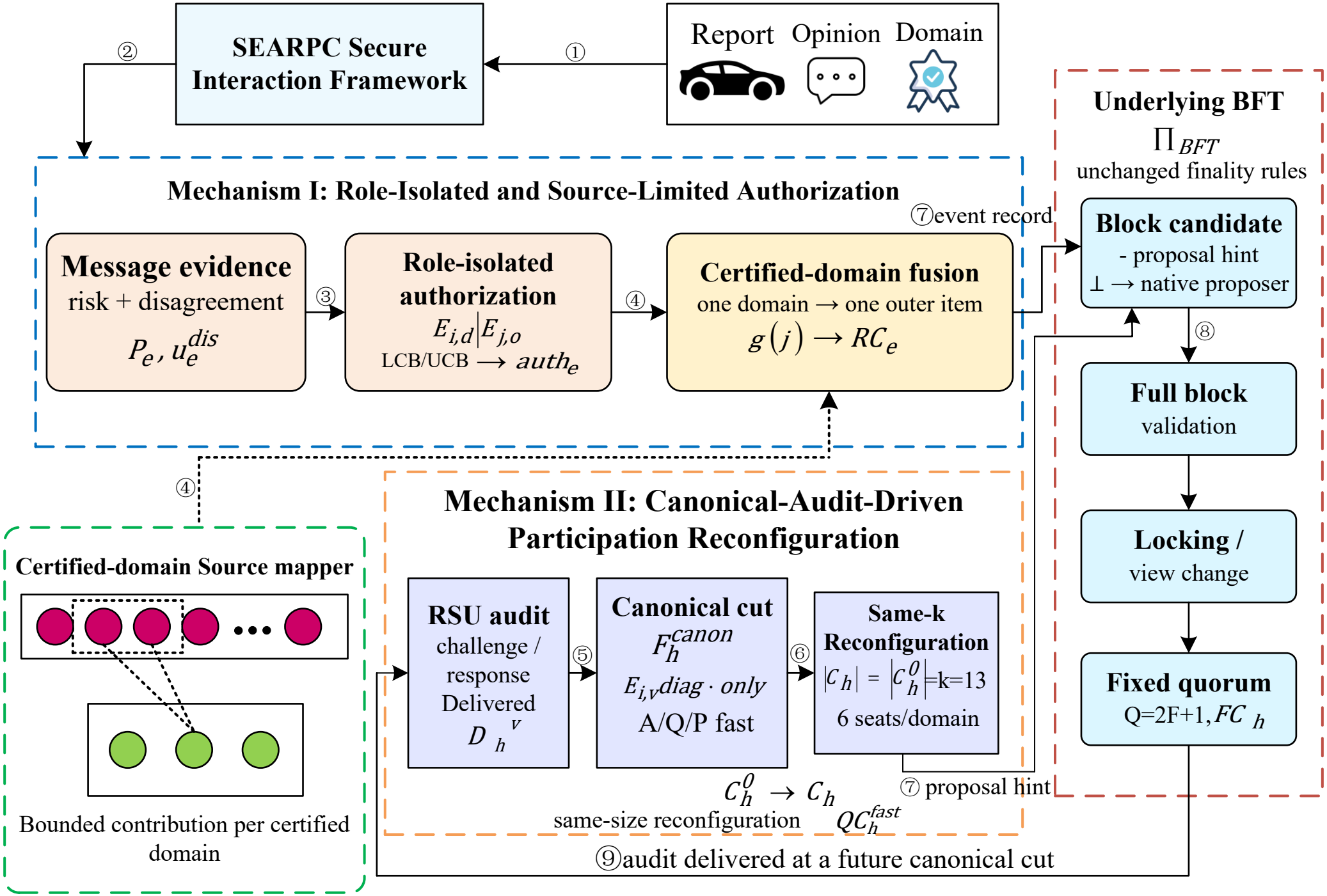

SEARPC Secure Interaction Framework
Report
Opinion
Domain
①
②
Underlying BFT
Π_{BFT}
unchanged finality rules
Mechanism I: Role-Isolated and Source-Limited Authorization
⑦event record
Message evidence
risk + disagreement
P_e, u_e^{dis}
③
Role-isolated authorization
$E_{i,d} | E_{j,o}$
LCB/UCB ⟶ $auth_e$
④
Certified-domain fusion
one domain ⟶ one outer item
$g(j) \longrightarrow RC_e$
Block candidate
- proposal hint
⊥ ⟶ native proposer
⑧
Full block
validation
Locking /
view change
Fixed quorum
Q=2F+1, FC_h
④
Certified-domain Source mapper
Bounded contribution per certified domain
Mechanism II: Canonical-Audit-Driven Participation Reconfiguration
RSU audit
challenge / response
Delivered D_h^v
⑤
Canonical cut
F_h^{canon}
$E_{i,v}diag \cdot only$
A/Q/P fast
⑥
Same-k Reconfiguration
$|C_h| = |C_h^0|$=k=13
6 seats/domain
⑦ proposal hint
$C_h^0 \rightarrow C_h$
same-size reconfiguration
QC_h^{fast}
⑨audit delivered at a future canonical cut